

RF Design Course 2026

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June 1, 2026

1 Task 6 and 7

1.1 Number 4

Table 1: Impedance and Discrepancy for S11 at 2.5 GHz

Type	Formula	Impedance $Z = R + jX$ (Ω)
Model (Blue)	$Z = Z_0 \times (0.150 - j0.359)$	$7.5 - j17.95$
Measurement (Red)	$Z = Z_0 \times (0.141 - j0.537)$	$7.05 - j26.85$
Discrepancy Δ	$\Delta R = 0.45 \Omega$, $\Delta X = 8.9 \Omega$	

Table 2: Impedance and Discrepancy for S22 at 2.5 GHz

Type	Formula	Impedance $Z = R + jX$ (Ω)
Model (Blue)	$Z = Z_0 \times (0.252 - j0.210)$	$12.6 - j10.5$
Measurement (Red)	$Z = Z_0 \times (0.417 - j0.911)$	$20.85 - j45.55$
Discrepancy Δ	$\Delta R = -8.25 \Omega$, $\Delta X = 35.05 \Omega$	

1.2 Number 6

Table 3: Impedance and Discrepancy for S11 at 500 MHz

Type	S-Parameter	Impedance $Z = R + jX$ (Ω)
Model (Blue)	$0.878 \angle -53.415^\circ$	$15.85 - j97.30$
Measurement (Red)	$0.962 \angle -42.838^\circ$	$7.30 - j127.10$
Discrepancy Δ	$ \Delta S11 = 0.084$, $\Delta \angle = 10.577^\circ$, $\Delta R = 8.55 \Omega$, $\Delta X = 29.80 \Omega$	

Table 4: Impedance and Discrepancy for S22 at 500 MHz

Type	S-Parameter	Impedance $Z = R + jX$ (Ω)
Model (Blue)	$0.579 \angle -41.239^\circ$	$71.55 - j82.25$
Measurement (Red)	$0.734 \angle -29.190^\circ$	$89.65 - j139.20$
Discrepancy Δ	$ \Delta S22 = 0.155$, $\Delta \angle = 12.049^\circ$, $\Delta R = -18.10 \Omega$, $\Delta X = 56.95 \Omega$	

At 500 MHz, the simulated and measured S11/S22 magnitudes and phases are relatively close, with only moderate phase deviations.

This shows that the small-signal model represents the low-frequency behavior reasonably well.

1.3 Number 7

Table 5: S-Parameter and Impedance for S11 at 14 GHz

Type	S-Parameter	Impedance $Z = R + jX$ (Ω)
Model (Blue)	0.757 $\angle$ 113.154°	9.85 + j32.10
Measurement (Red)	0.813 $\angle$ - 166.376°	5.25 - j5.90
Discrepancy Δ	$ \Delta S_{11} = 0.056$, $\Delta \angle = 279.53^\circ$, $\Delta R = 4.60 \Omega$, $\Delta X = 38.00 \Omega$	

Table 6: S-Parameter and Impedance for S22 at 14 GHz

Type	S-Parameter	Impedance $Z = R + jX$ (Ω)
Model (Blue)	0.872 $\angle$ 48.535°	19.80 + j107.90
Measurement (Red)	0.856 $\angle$ - 155.073°	4.10 - j11.00
Discrepancy Δ	$ \Delta S_{22} = 0.016$, $\Delta \angle = 203.61^\circ$, $\Delta R = 15.70 \Omega$, $\Delta X = 118.90 \Omega$	

At 14 GHz, large deviations between the simulated and measured S11/S22 can be observed in both magnitude and phase.

This indicates that the model cannot accurately capture high-frequency parasitic effects.

1.4 Number 8

Table 7: S-Parameter Magnitudes and Ratio at 14 GHz

Quantity	Model (Blue)	Measurement (Red)
S_{21} (dB)	-8.533	-10.415
$ S_{21} $ (linear)	0.375	0.301
S_{12} (dB)	-8.089	-34.152
$ S_{12} $ (linear)	0.394	0.0197
$ S_{21} / S_{12} $	0.95	15.3

The measured ratio $|S_{21}|/|S_{12}|$ is about 15.3, showing a clear forward gain and reverse isolation in the device.

The model fails to capture this isolation.

1.5 Number 9

Note: Please optimize the design "AfterOpt" to see the optimized result.

S-Parameters Comparisons After Optimization AmodelB

Blue Trace : Simulation (Model) Data

Red Trace : Measured Data

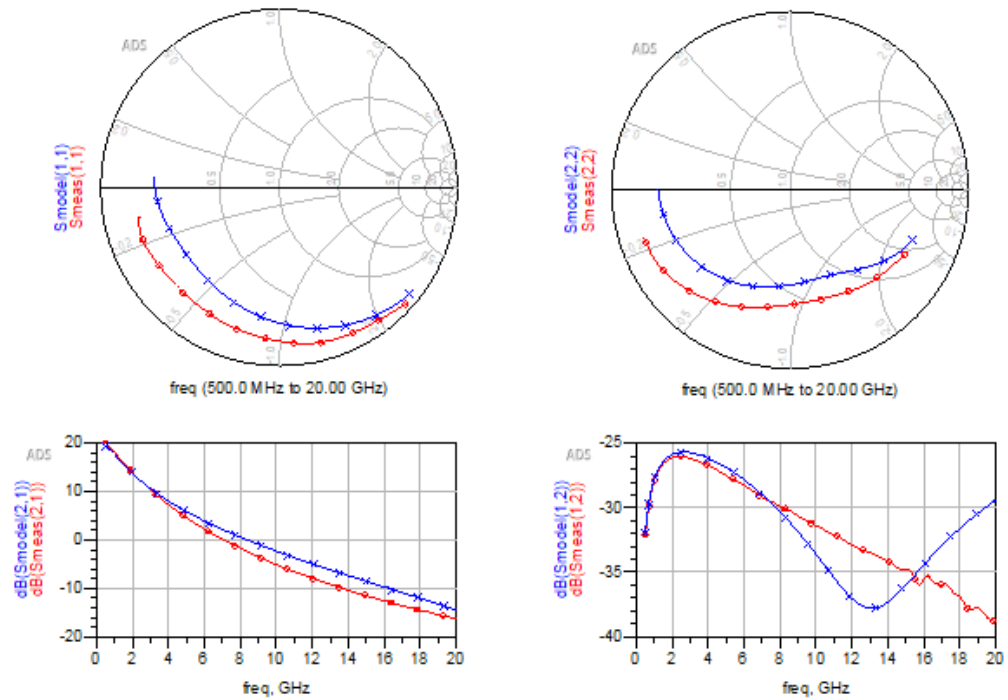


Figure 1: Model isolation analysis response

1.6 Number 10

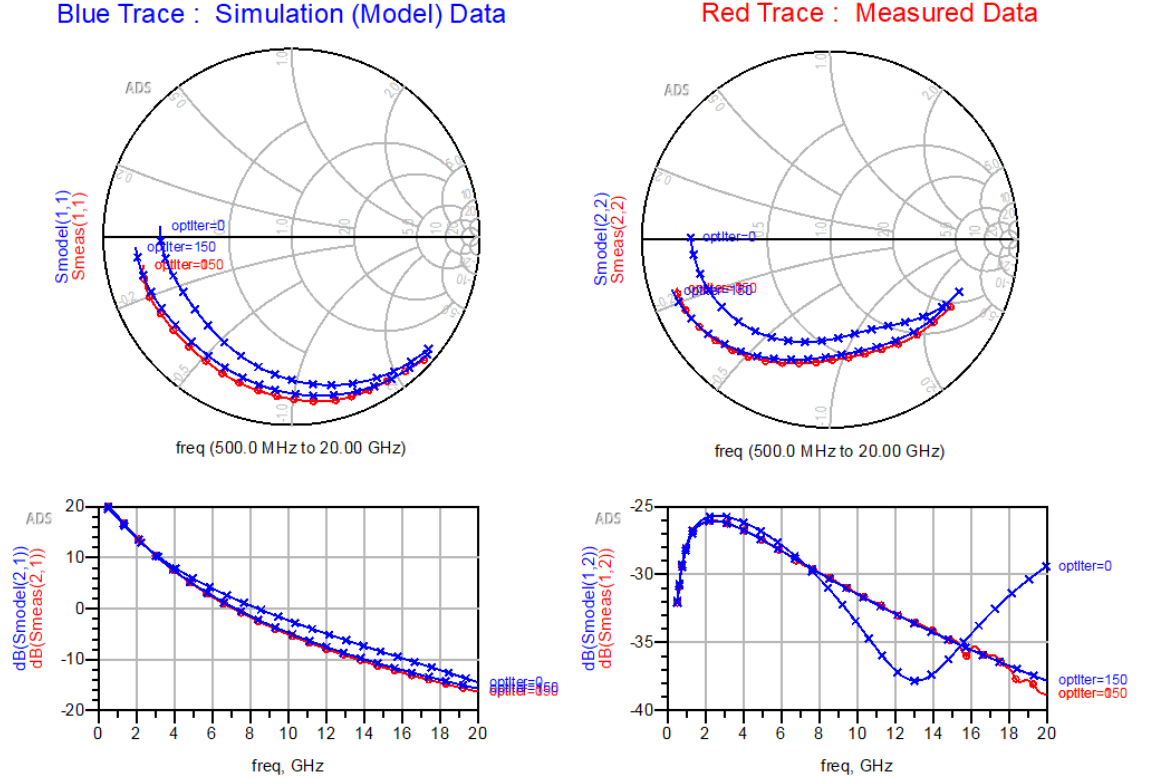


Figure 2: Optimizer Response for 150 steps

1.7 Number 11

1. **Objective Minimization:** The optimizer iteratively adjusts the variables of the small-signal model to minimize a Goal (Error Function), which is the weighted difference between the simulated and measured S-parameters.
2. **Local Minima:** Optimization often stalls because the algorithm gets stuck in a “local minimum,” where any small change in parameters increases the error, even if a much better global solution exists elsewhere.
3. **Model Topology Limitations:** If the chosen equivalent circuit is too simple and fails to account for real-world physical parasitics (like pad capacitances or substrate effects), it is mathematically impossible for the model to perfectly replicate the measured data.

4. **Parameter Trade-offs:** Transistor parameters are often interdependent; the optimizer must frequently compromise the accuracy of one S-parameter (e.g., S_{22}) to improve the fit of another (e.g., S_{11} or S_{21}).
5. **Data Inconsistencies:** Discrepancies persist because the optimizer tries to fit an ideal mathematical model to real-world measurements that contain measurement noise or calibration errors, especially at higher frequencies.

1.8 Number 12

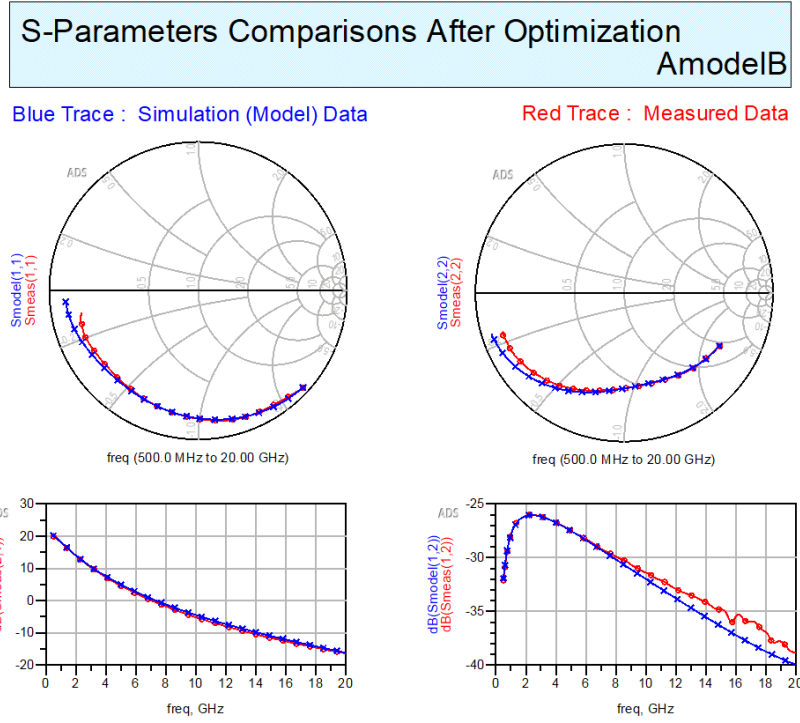


Figure 3: Optimized in between 0.5 to 2 GHz with S-Parameter re-simulated.

1.9 Number 13

Up to 9 GHz, the magnitude and phase agree reasonably well between the measurement and the optimized model.

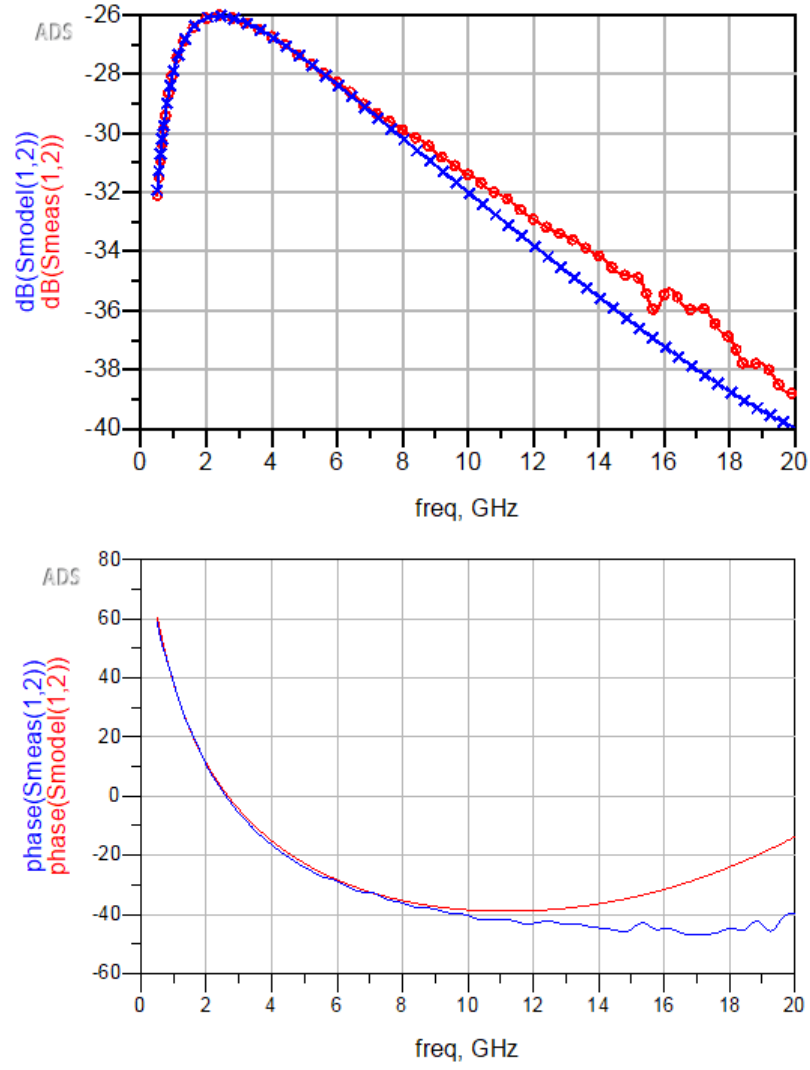


Figure 4: S_{12} Magnitude and Phase Plot.

1.10 Number 16

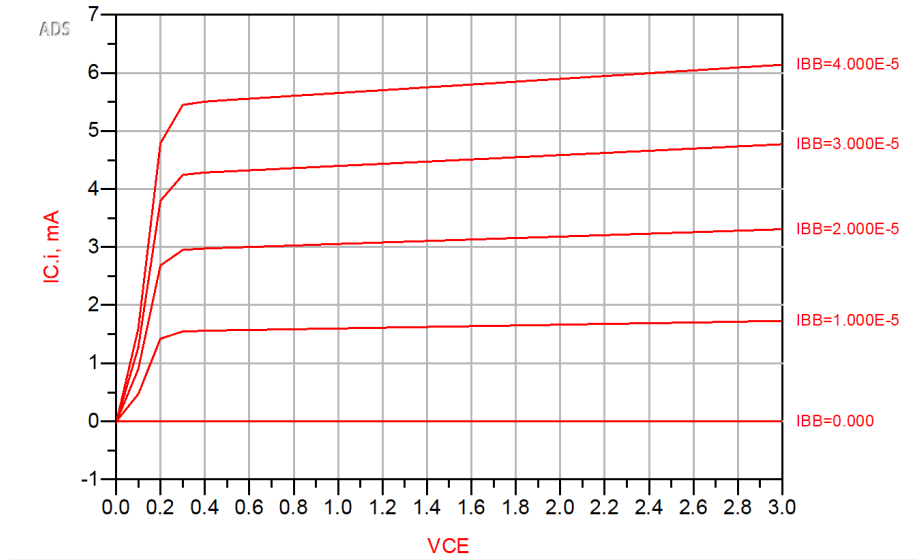


Figure 5: BJT Output Curve - ICE

1. At high V_{CE} and high current levels, real bipolar transistors experience avalanche breakdown (impact ionization) in the base-collector junction, causing the collector current to rise sharply.
2. Thermal runaway or self-heating can occur because the high power dissipation ($P = V_{CE} \cdot I_{CE}$) increases the device temperature, which inherently boosts minority carrier injection and further amplifies the current.

1.11 Number 17

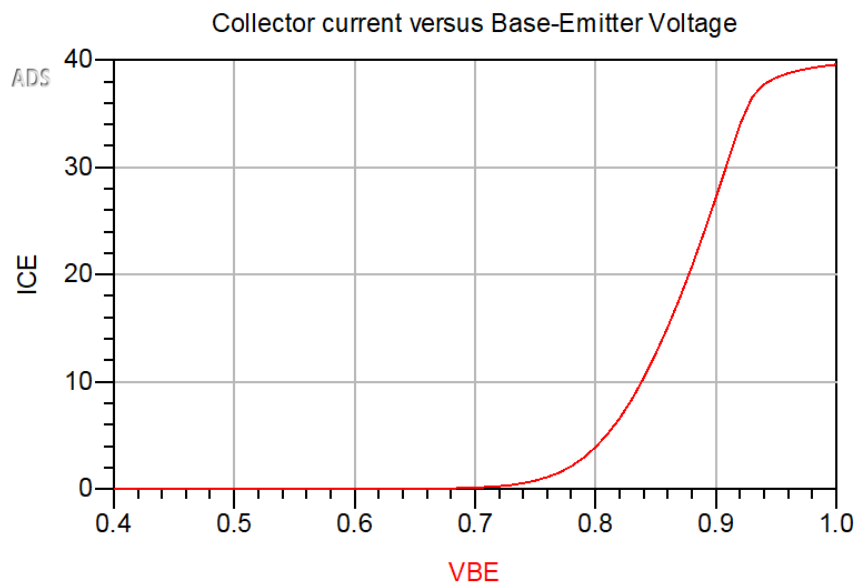


Figure 6: BJT characteristic curve: I_{CE} vs. V_{BE} .

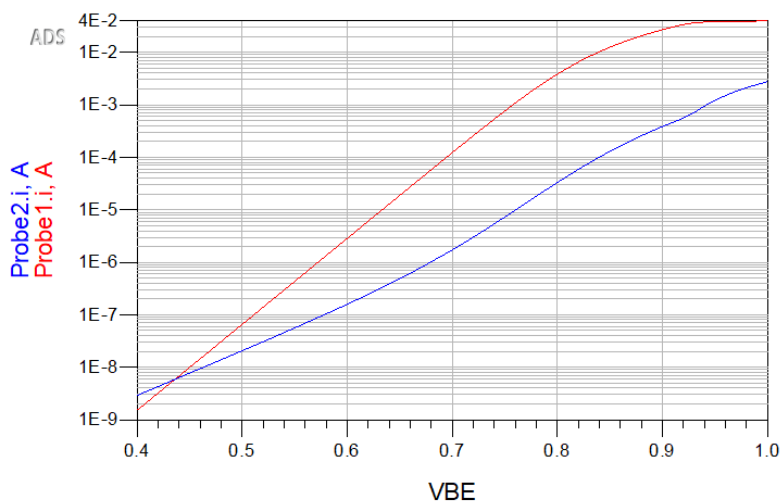


Figure 7: Gummel Plot: I_C (Probe 1) and I_B (Probe 2) vs. V_{BE} on a logarithmic scale.

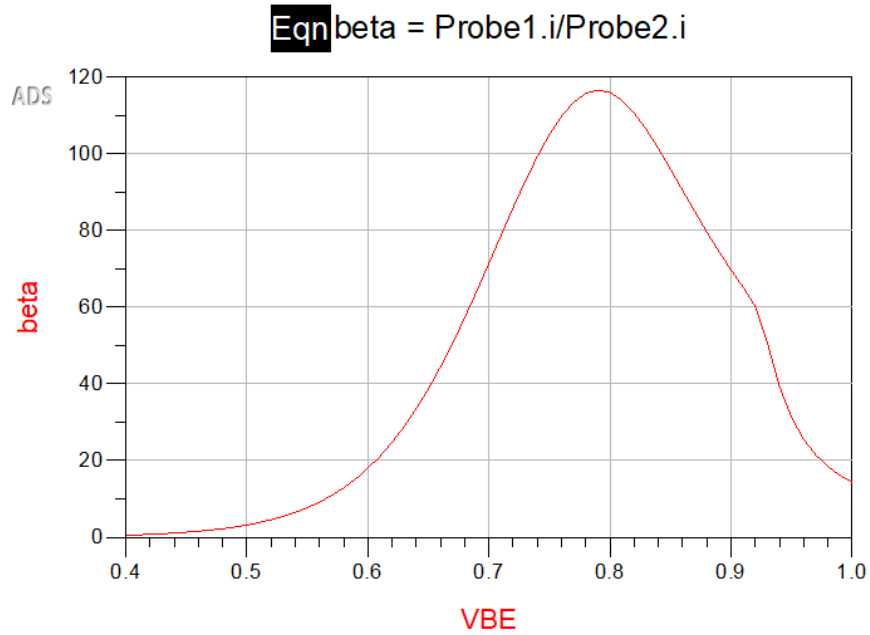
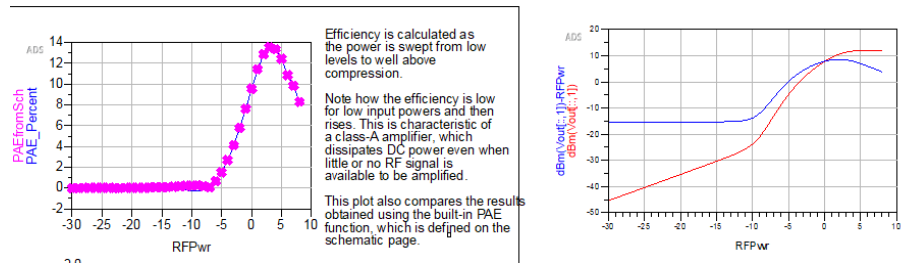


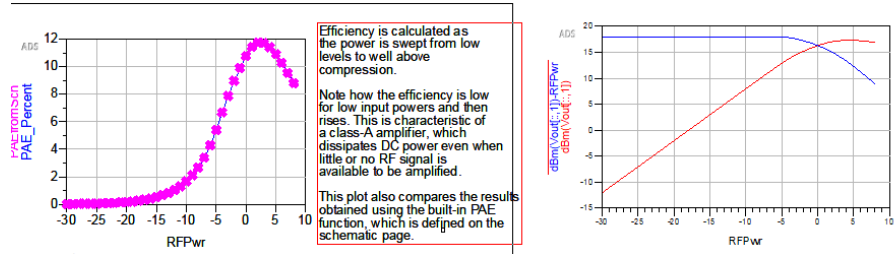
Figure 8: Common-emitter current gain (β) as a function of the I_C/I_B ratio.

2 Task 8

2.1 Number 2

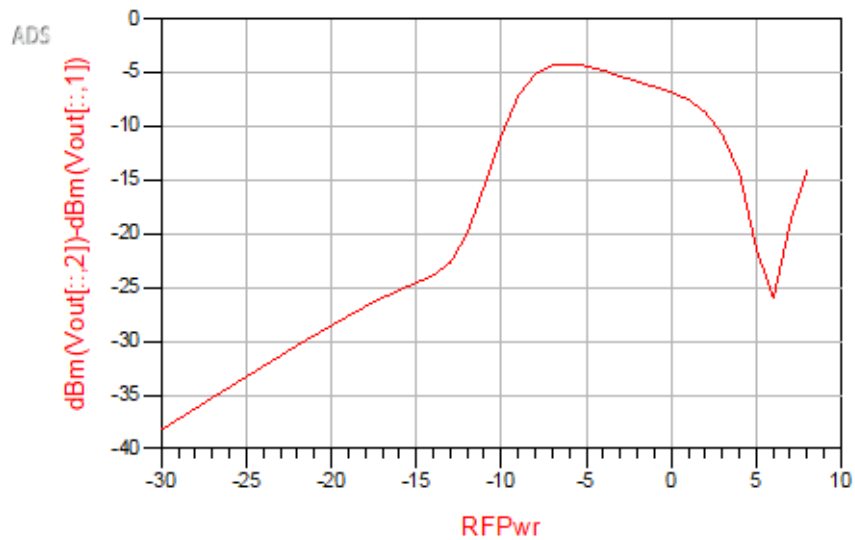


2.2 Number 3

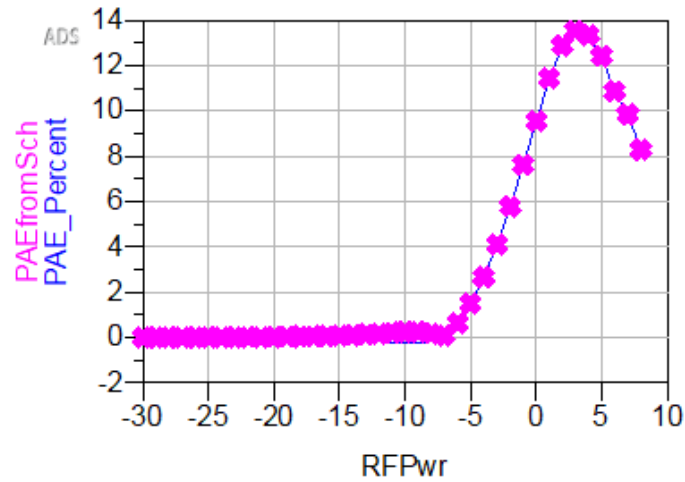


- At $V_{DS} = 12\text{ V}$, the amplifier has a higher small-signal gain and a wider linear range than at 6 V.
- The saturated output power is significantly increased at 12 V due to the larger available output voltage swing.
- Ideally, doubling the supply voltage can boost the saturated output power by up to a factor of 4 (+6 dB).

2.3 Number 4

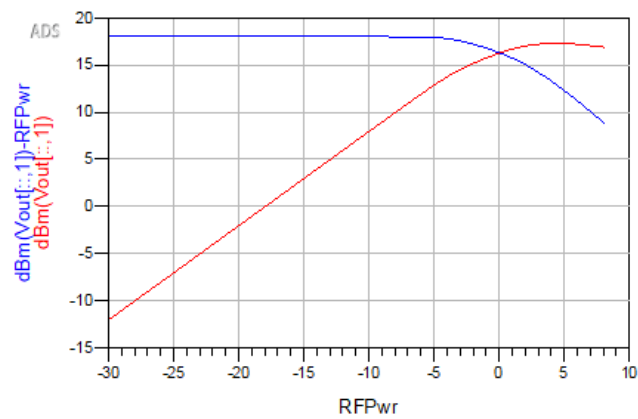


2.4 Number 5

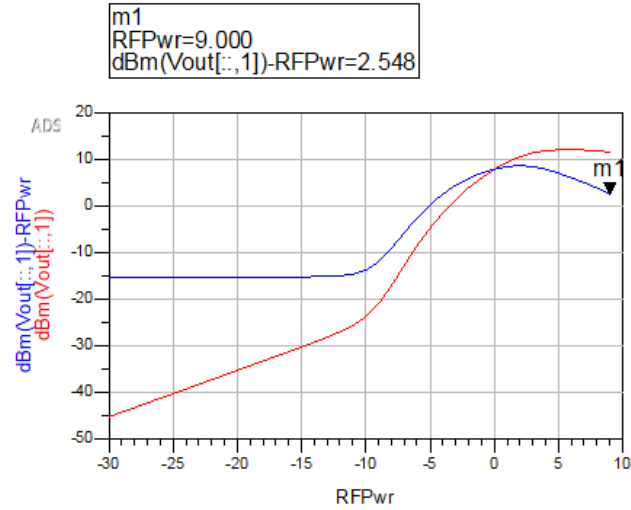


- At high compression, the PAE peaks and then decreases as the amplifier saturates.
- The drop occurs because output power growth slows while DC power consumption rises.
- This reduces the overall efficiency of DC-to-RF power conversion.

2.5 Number 6

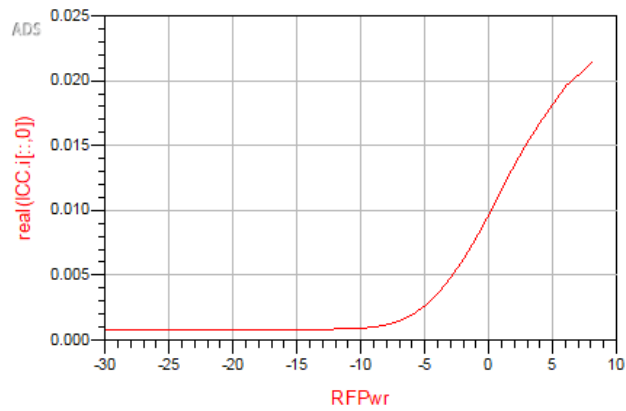


2.6 Number 7



At $V_{DS} = 6$ V and input power of 9 dBm, the gain is compressed by 2.548 dB compared to its small-signal linear gain.

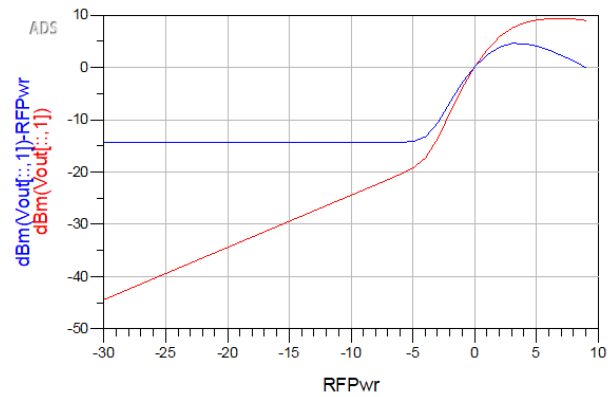
2.7 Number 8



- At low input power, the DC bias current remains constant at its quiescent value.
- As the amplifier enters compression, the average DC current rises significantly due to a wider conduction angle.

2.8 Number 9

Vdc=4V



1. The small-signal gain drops noticeably when supply voltage falls below 6 V.
2. Gain compression starts at a much lower input power level.
3. The maximum saturated output power is reduced significantly.

3 Task 9

3.1 Number 3

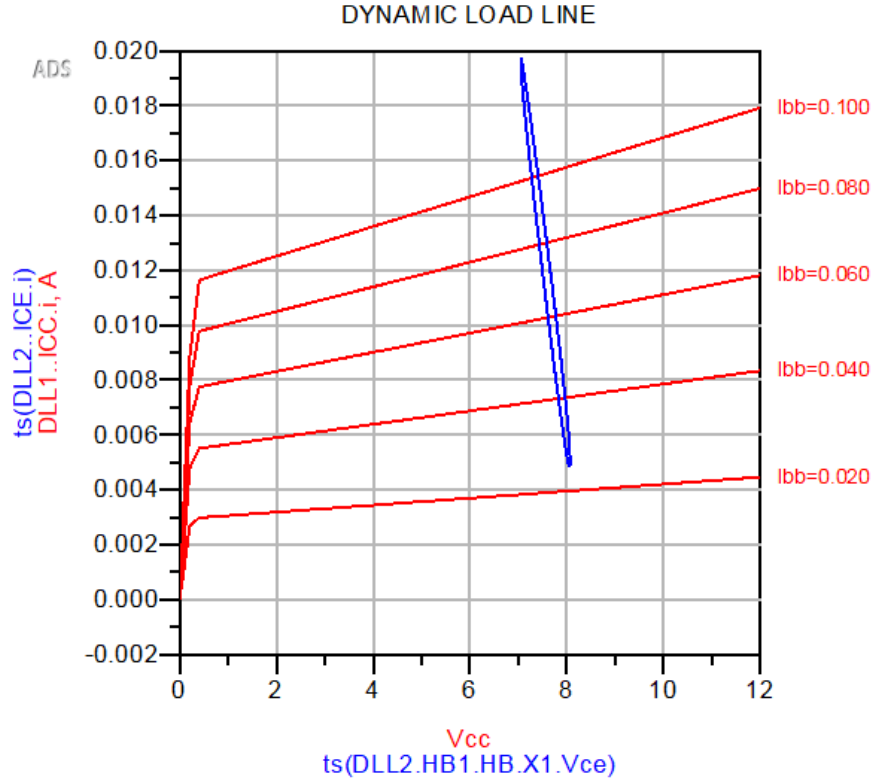


Figure 9: Loadline Plot

1. The ideal positioning of the dynamic load line maximizes both the output voltage and current swings without driving the transistor into hard clipping or dielectric breakdown.
2. Centered symmetrically around the current 10 V DC bias point, the trajectory ensures that the positive and negative cycles of the RF signal expand evenly to minimize premature harmonic distortion.
3. On the left boundary, the peak RF current should reach its maximum value (I_{max}) precisely at the edge of the knee voltage region to maximize efficiency while avoiding linear gain degradation.
4. On the right boundary, the peak RF voltage must swing as close to the drain breakdown voltage limit as safely possible at zero current.
5. This specific alignment ensures the maximal utilization of the transistor's

boundary area, thereby achieving the highest possible output power and efficiency.

3.2 Number 4

1. Choosing an even higher supply voltage beyond the baseline 10 V increases the available voltage headroom, which fundamentally scales up the required optimum large-signal load resistance value when expressed in Ohms ($R_{\text{opt}} \approx \frac{V_{\text{max}} - V_{\text{knee}}}{I_{\text{max}}}$).
2. Because the maximum current capability (I_{max}) remains physically fixed at roughly 110 mA by the transistor's upper saturation boundaries, the load impedance must increase to a higher resistance value to accommodate this wider voltage swing.
3. Consequently, since resistance is inversely proportional to the slope on an I - V plot, this higher value in Ohms causes the real part of the ideal load line to become noticeably flatter and less steep.